🎓 Campusly Marketplace — Comprehensive System Overview

> [!NOTE] **Campusly** is an open, modern, high-performance student commerce ecosystem designed for campus communities. It enables students to freely list, buy, sell, and trade any legal product, digital download, freelance service, or community access pass.

📄 PAGE 1: System Architecture & Technology Stack

🏗️ 1. High-Level Architecture Diagram

```
graph TD
    subgraph Client Layer ["Client Layer (Next.js 16 App
        PublicMP["Public Marketplace\n(/, /product/[id],
        Dashboard["Student Seller Dashboard\n(/dashboard,
        AdminPortal["Admin Control Center\n(/admin, /veri
    end

    subgraph API Layer ["API & Core Services (NestJS Engi
        AuthModule["Auth Module\n(JWT, RBAC Guards, Bcryp
        ProductsModule["Products Module\n(Moderation Stat
        AdminModule["Admin Engine\n(Analytics, Verificati
        TxModule["Transactions Engine\n(Sales & Checkout)
        UploadsModule["Uploads & Storage\n(Cloudinary Ser
    end

    subgraph Data Layer ["Persistence & Infrastructure"]
```

```
    Postgres["PostgreSQL Database\n(Prisma ORM Schema
    Cloudinary["Cloudinary Storage\n(ID Cards, Images
end

Client Layer -->|REST API + JWT Bearer| API Layer
AuthModule --> Postgres
ProductsModule --> Postgres
AdminModule --> Postgres
TxModule --> Postgres
UploadsModule --> Cloudinary
```

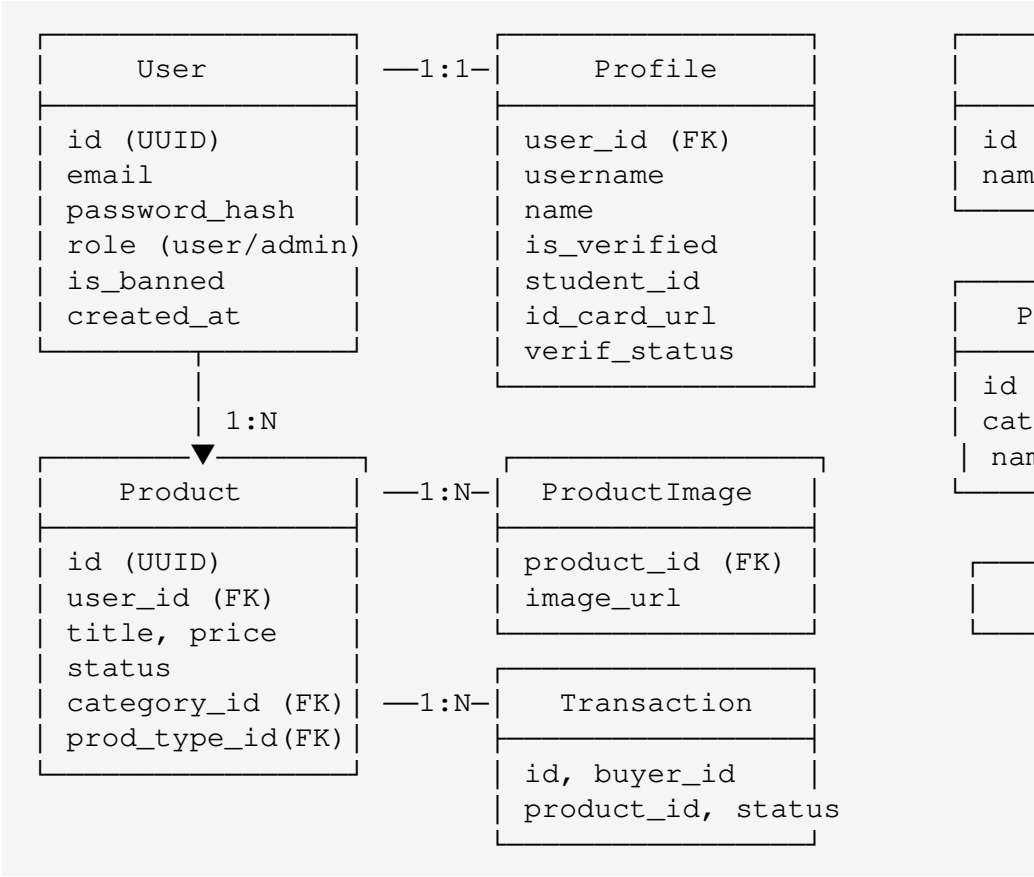
💻 2. Technology Stack & Framework Matrix

Layer	Technology	Primary Function & Responsibilities
Frontend Framework	**Next.js 16 (App Router)**	Server & Client Components, Turbopack bundling, dynamic routing
User Interface	**Tailwind CSS + Lucide Icons**	Glassmorphism aesthetics, dark/light modes, micro-animations
State & Auth	**React Context + LocalStorage**	`AuthContext` token lifecycle, platform config sync
Backend Framework	**NestJS (TypeScript)**	Modular Controller-Service architecture, Dependency Injection
Database & ORM	**PostgreSQL + Prisma ORM**	Relational schemas, transactions, migrations, relational queries
Authentication	**JWT + Passport + Bcrypt**	Bearer authentication tokens, role-based guards (`user` / `admin`)

File Storage	**Cloudinary API Integration**	Secure uploads for Student IDs, product images, and deliverables

3. Core Database Models (Prisma Schema)

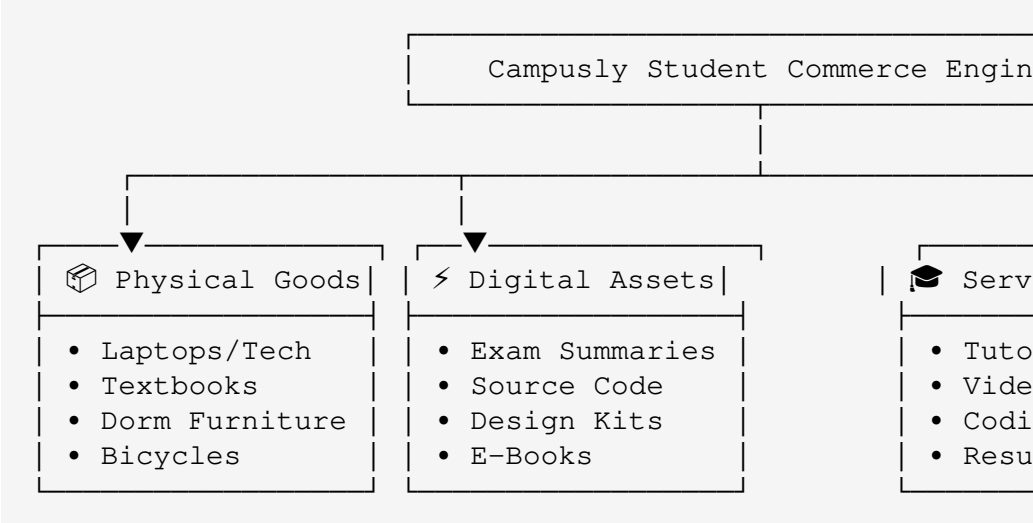
> [!TIP] The relational schema connects Users, Profiles, Products, Product Categories, Product Types, Product Reviews, and Transactions.



📄 PAGE 2: Open Commerce Engine, Capabilities & Admin Portal

🛍️ 1. Multi-Format Open Commerce Engine

> [!IMPORTANT] Unlike single-purpose academic repositories, Campusly is engineered to support **four distinct listing types**, allowing students to monetize any legal product or service.



Key Commerce Capabilities:

1. **Instant Digital Deliverables**: Buyers automatically gain access to deliverable files/links immediately after checkout.
2. **Instant vs. Moderated Publishing**: Configurable auto-approval engine lets admins toggle whether user listings go live instantly (⚡ `Instant Publishing`) or undergo manual review.
3. **Student Identity Verification (Blue Tick)**: Verification portal allows students to upload Student IDs to earn verified seller trust

badges.

🛡️ 2. Admin Control Center & Operations Ecosystem

The Admin Portal (`/admin`) provides full administrative oversight and platform governance:

Admin Module	Route	Capabilities & Features
Overview Analytics	`/admin/overview`	Platform revenue, active listings, transaction totals, daily timeseries charts
Pending Queue	`/admin`	Review pending listing submissions, approve or request revisions with reviewer notes
Content Monitor	`/admin/activity`	Monitor active marketplace listings, audit reported content, force-delete policy violations
User Management	`/admin/users`	View all accounts, inspect seller details, issue bans with mandatory audit reasons, unban users
Verification Portal	`/admin/verifications`	Review uploaded student ID documents, issue blue ticks, or reject applications with feedback
Support & Messages	`/admin/messages`	Unified message center for verifications, listing reviews, completed sales, policy reports, and direct user notices

Taxonomy & Settings	`/admin/settings`	Tree-view category & sub-category editor, auto-approval moderation toggle, admin PFP & credential manager

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🚀 3. Security & Platform Guardrails

> [!WARNING] Campusly strictly enforces platform policies and secure operational standards.

- **Role-Based Access Control (RBAC)**: All `/admin/*` routes and API endpoints are guarded with NestJS `JwtAuthGuard` & `RolesGuard('admin')`.
- **Proactive Content Protection**: Buyers can report non-compliant content, automatically routing items to the Admin Content Monitor.
- **Auditability**: Every administrative action (bans, listing rejections, verification decisions) is logged with administrative notes.

📌 Summary

Campusly merges modern web design aesthetics (**Next.js 16, Tailwind, Turbopack**) with a resilient backend architecture (**NestJS, PostgreSQL, Prisma, Cloudinary**), delivering a fast, secure, and infinitely scalable marketplace for campus communities.